Andrew Taggart

Software Engineer – Data Science – Machine Learning

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SKILLS

Languages: Python, R, JavaScript, HTML5, CSS3, SQL, PHP | **Frameworks:** Spark, Vite, React, Node.js, Express.js, Django | **Libraries:** pandas, matplotlib, seaborn, plotly, NumPy, scikit-learn, PyTorch, SHAP, leaflet | **Databases:** MongoDB, PostgreSQL | **Tools:** Github, VSCode, Jupyter, Figma, Microsoft Office, JSON, Tableau, Docker | **Skills:** REST APIs, ETL, EDA, CRUD

RESEARCH & PROJECTS

Federated Healthcare Explainability Research

- Developed federated learning experiments using PyTorch and FedAvg-style aggregation to evaluate model performance and explainability across 2-client and 5-client settings for classification and survival prediction tasks.
- Built preprocessing and evaluation pipelines with pandas, NumPy, and scikit-learn, including stratified splits, one-hot encoding, standardization, Macro-F1, C-index, loss tracking, and early stopping.
- Implemented model explainability analysis using permutation importance and SHAP to compare feature-attribution stability across centralized, global federated, and local client models.

NJ-EQUIP: Spatiotemporal Analytics and Visualization Portal for Energy Equity

- Led development of an interactive energy-equity web portal that simplified environmental, utility, and demographic datasets into accessible visual tools.
- Built React frontend features including routed page views, layered geographic map interfaces, responsive UI components, and an energy burden estimator using utility costs, heating costs, and household income.
- Supported paper and website revisions with the lead and co-author; project awarded Best Paper at IEEE CCWC 2026.

Venue Purchase Portal (Ticketly) | (React.js, Python, PostgreSQL)

- Led backend development and integration with the frontend, ensuring cohesive functionality.
- Implemented and optimized the Event Browsing and Venue Discovery features, providing users with an intuitive interface to explore various events and venues, contributing to a seamless user experience.
- Collaborated closely with team members on database design and frontend development, ensuring efficient data management and a cohesive, responsive UI across all pages, including Home, Venue, Events, and Reviews.

EXPERIENCE

Body Fit Training | Coach, Montclair & Hoboken, New Jersey — May 2024 – April 2025

Research based group fitness coaching with a focus on strength training.

- Built rapport with clients by creating a welcoming, high-energy experience for members throughout each session.
- Provided real-time coaching, movement corrections, and exercise modifications based on each member's individual needs.
- Supported member engagement and retention by maintaining a positive training atmosphere and professional client interactions.

Equinox | Personal Trainer**,** New York, NY — October 2022 - December 2023

Results-driven sales and program management of luxury fitness experience

- Communicated the value of personal training, driving sales.
- Educated clients on healthy lifestyle modifications, proper technique, and safety precautions.
- Established strong relationships with clients based on trust, respect, and professionalism.

Milwaukee Tools (TTI) | Global Electronics Sourcing Analyst, Brookfield, Wisconsin — July 2021 – July 2022

Collection and analysis of data on semiconductors and relative projects

- Analyze forecasts of semiconductor cost and usage to communicate needs throughout the supply chain.
- Initiate and maintain positive relationships with semiconductor suppliers.
- Cooperate with global sourcing and procurement teams to maintain flow of supply information.
- Facilitate ongoing and new engineering projects needs for semiconductor demand.

Techtronic Industries Co. (TTI) | Field Sales and Marketing Representative**,** Delray Beach, Florida — July 2020 - June 2021

Managed on-site sales and marketing efforts in partnered stores.

- Maximized sell through of brands by analyzing market potential and properly merchandising products.
- Strengthened and maintained positive relationships with customers, TTI Management, and our retail partners.
- Monitored sales and analyzed monthly results to identify additional sales opportunities.
- Organized in-store product training sessions and demonstrations to educate store staff and customers.

EDUCATION

Montclair State University | M.S., Data Science | Montclair, NJ | **GPA:** 3.874 — **May 2026**

General Assembly | *Software Engineering Immersive* | Remote — **April 2024**

University of Miami | B.S., *Human and Social Development* | Coral Gables, FL — **May 2020**